Project Summary:

This project successfully implemented a scalable and auditable **Extract, Load, Transform (ELT)** pipeline using **dbt** (data build tool) within a **Snowflake** data warehouse environment. The core objective was to transform granular, time-series stock exchange data from the raw landing zone into a highly structured **Star Schema** . This architecture ensures data quality through rigorous **Staging** (cleaning and standardization) and **Testing** (validation of primary keys and non-null constraints), resulting in clean analytical assets like FACT_STOCK_PRICES and DIM_DATE. The final data structure facilitates efficient analytical querying and is directly consumed by a professional Power BI dashboard, effectively converting complex financial time-series data into actionable business intelligence.

Daily-Life Use Cases

- Leveraging a data warehouse such as yours (built for PSX), you — or others — could use it for a variety of practical, real-life purposes:

- Historical trend analysis for investments — e.g. evaluating long-term performance of stocks, comparing past and current behaviour to inform buy/sell decisions. A warehouse makes it easier to query years of data quickly.

- Portfolio performance reporting / personal finance dashboards — building dashboards that show returns, volatility, sector-wise exposure over time, allowing individual or institutional investors to monitor and manage portfolios.

- Market-wide analytics & pattern detection — analyzing aggregate metrics: e.g. tracking the most traded sectors over time, average trading volumes or volatility patterns, helping a trader or analyst identify macro trends or market cycles.

- Backtesting investment strategies / simulating scenarios — by using historical data to simulate how certain trading strategies would have performed in the past, which is crucial for risk management or algorithmic trading.

- Regulatory or compliance reporting / auditing — an aggregated warehouse facilitates generating reports for regulatory compliance, tax or audit purposes, ensuring all historical transactions are stored in consistent format and traceable.

- Research and academic studies — enabling students, economists, or researchers to run statistical analyses on the PSX data: study correlations, market behaviour before/after events, volatility clustering, etc.

- Investor education tools — building educational dashboards or tools for new investors to learn from historical data — e.g. illustrating how market crashes or booms evolved, how different stocks or sectors performed, etc.

These use-cases reflect common advantages of data warehouses: consolidated data from multiple periods, data cleansing to ensure quality, fast querying for analytics, and support for reporting or decision-making.

Reference Link:

https://www.sap.com/westbalkans/products/data-cloud/datasphere/what-is-a-data-warehouse.html?utm_source=chatgpt.com

https://www.freshconsulting.com/insights/blog/data-warehouse-development-a-guide-and-case-study/?utm_source=chatgpt.com